

Gopal Kumar Jha

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SOFTWARE ENGINEER

I am a final-year B.Tech student with a strong interest in distributed systems, concurrency, multithreading, synchronization, computer networks, debugging and system design. I actively study and build my understanding of artificial intelligence, machine learning, and data mining.

TECHNICAL SKILLS

Languages : C/C++, Java, Python, JavaScript, TypeScript, Go, SQL

TechnologiesTool: Spring Boot, Node.js, Express, Git, GitHub, Docker, Kubernetes, Kafka, PostgreSQL, Redis.

PROJECTS

Second year

Multi-Threaded HTTP File Server

- Developed a multi-threaded HTTP file server to enable secure and concurrent file sharing for multiple users without performance bottlenecks.
- Implemented thread pooling, synchronization, and authentication to ensure efficient request processing and reliable access to shared resources.
- Designed RESTful APIs for file upload, download, and management with persistent metadata storage.
- Used: Java, Spring Boot, Java Concurrency (ExecutorService), PostgreSQL, Docker, Postman.

Second year

Smart Network Traffic Monitor

- Built a network traffic monitoring system to analyze real-time network activity and identify suspicious traffic patterns for improved network visibility.
- Captured and processed network packets, generated traffic statistics, and detected common anomalies such as port scans and unusual bandwidth usage.
- Developed dashboards and reporting features to support network monitoring and troubleshooting.
- Technologies: Python, Scapy, Wireshark, Pandas, Matplotlib.

Third year

Clustering SSH Attacks

- Developed a machine learning-based network security solution to identify and group similar SSH attack patterns using network traffic analysis.
- Processed PCAP files, extracted traffic features, and applied K-Means clustering to detect anomalous SSH behavior.
- Evaluated clustering quality and visualized attack patterns to support security analysis.
- Technologies: Python, Scikit-learn, Pandas, NumPy, Scapy, Matplotlib, Git.

Third year

URL Shortener with Analytics

- Developed a scalable URL shortening service to simplify link sharing and provide real-time analytics on link usage.
- Implemented unique short URL generation, caching, click tracking, and rate limiting for efficient request handling.
- Designed REST APIs and optimized database queries for fast URL redirection under concurrent user requests.
- Technologies: Go, Gin, PostgreSQL, Redis, Docker Postman.

EDUCATION

Quantum University

Bachelor of Technology

Aug 2023 – June 2027

CGPA: 6.88/10

Relevant coursework: Object Oriented Programming, Databases, Discrete Maths, Data Structures and Algorithms, Operating Systems, Computer Networks, Advance Data Structures and Algorithms, Information Retrieval, Image Processing